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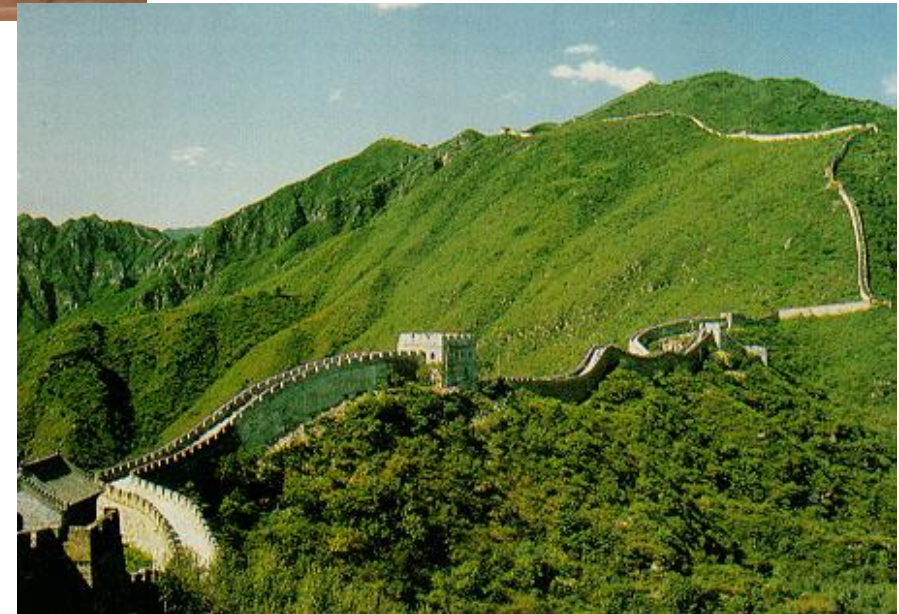
CH0: INTRODUCTION

9 FEBRUARY 2026

Project Management 412

Prof Corné Schutte

Evidence of projects is everywhere...



Recent Examples

Millennium Park, Chicago

- Groundbreaking targets, 1998:
 - Total cost: \$150 million
 - Gehry band shell: \$10.8 million
 - Completion: 2000 (millennium!)
- Actual
 - Total Cost: \$475 million
 - Gehry band shell: \$60.3 million
 - Completion date: Summer 2004



Recent Examples

Greenpoint (DHL) Stadium – 2010 World Cup

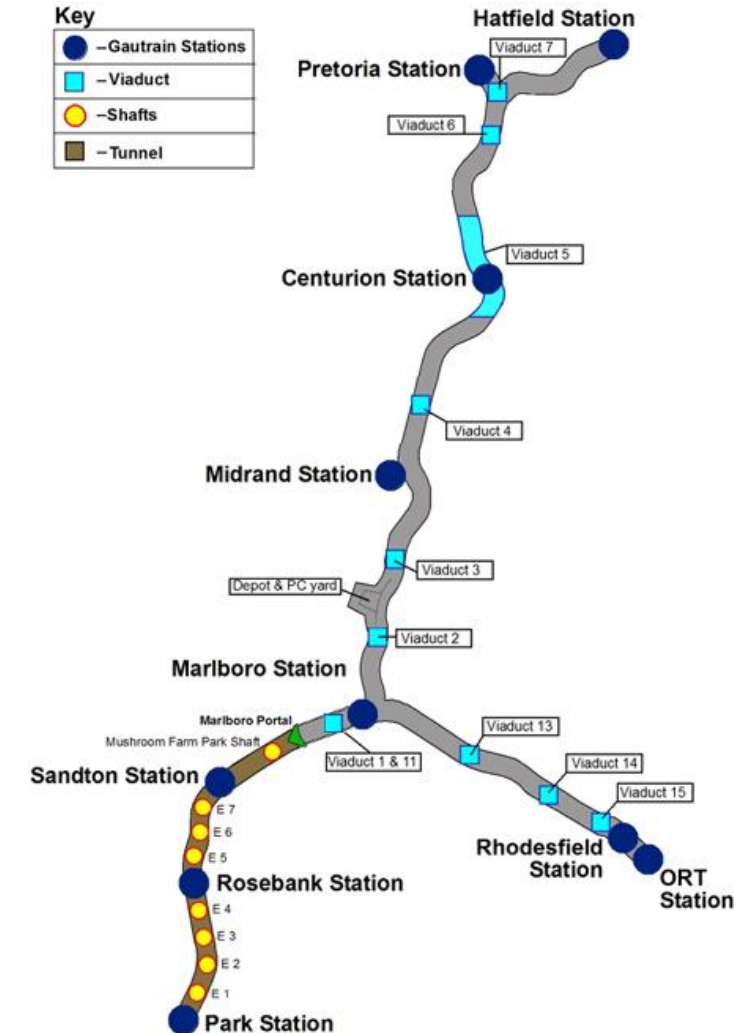
- Initial estimate (2006): R1.2 B
- Tender price M&R & WBHO (2007): R2.85 B
- Final price (2009): R4.4 B



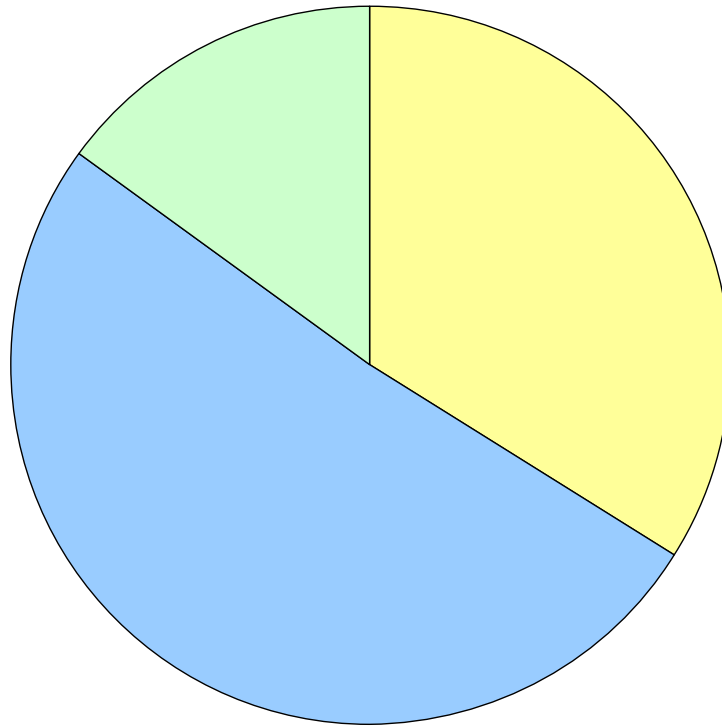
Recent Examples

Gautrain

- Initial Concept (2000–2002): R3.5 billion to R4 billion.
- Pre-Construction (2003–2005): R7 billion in 2003, R20 billion by 2005.
- Approved Budget (2006): Bombela Concession Company R25.2 billion.
- Final Construction Cost (2011–2012): R26.2 billion.



Successful, Challenged, or Failed?



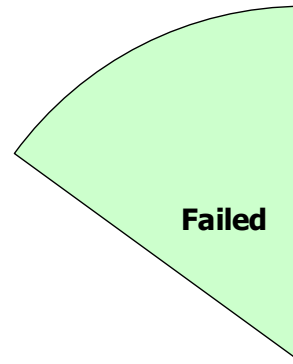
Success = On time, within budget, committed scope

Challenged = Late, overbudget, or lacking key scope

Failed = Canceled before completion or never implemented

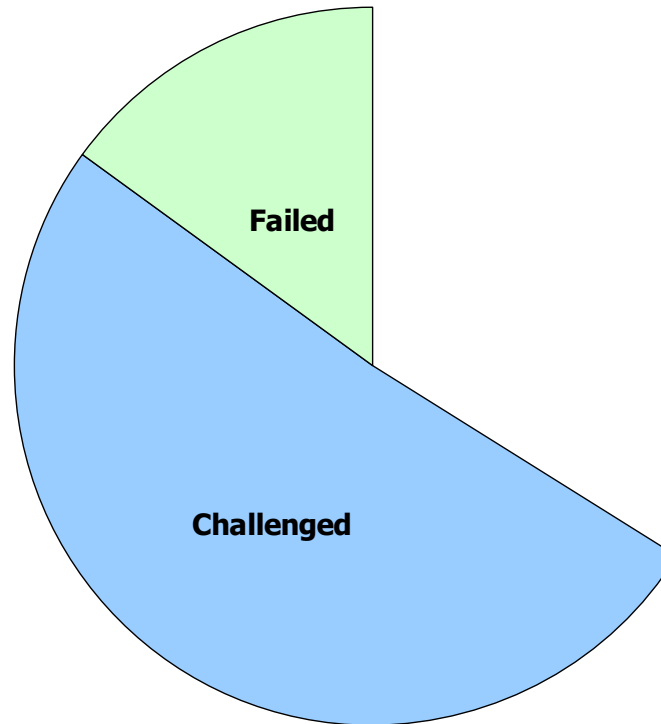
From 2004 Standish report

The Answers



15% of projects were cancelled before they finished

The Answers



51% of projects were late, over-budget, or lacked scope (probably due to schedule challenges)

The Answers



Only 31% of projects finished on-time, within budget, and with committed scope

Performance of IT Projects



2003 “Chaos” Report, Standish Consulting Group

- Major IT projects that fail, 66%
- Average cost overrun, 43%
- Projects with schedule overrun, 82%
- Required features/functions not included in released system, 48%

Criteria for Project “Failure” or “Overrun”

- >20% over budget, and
- >20% late, and
- >20% of business requirements not met

Why do projects fail or overrun schedules/budgets?



Typical reasons

- Weather
- Inadequate requirements definition
- Insufficient resources
- Changing priorities of customer or management
- Intractable technical problems
- Resistance from stakeholders
- Wrong project for the stated needs
- Inadequate tracking and control
- Inexperienced project manager and/or team

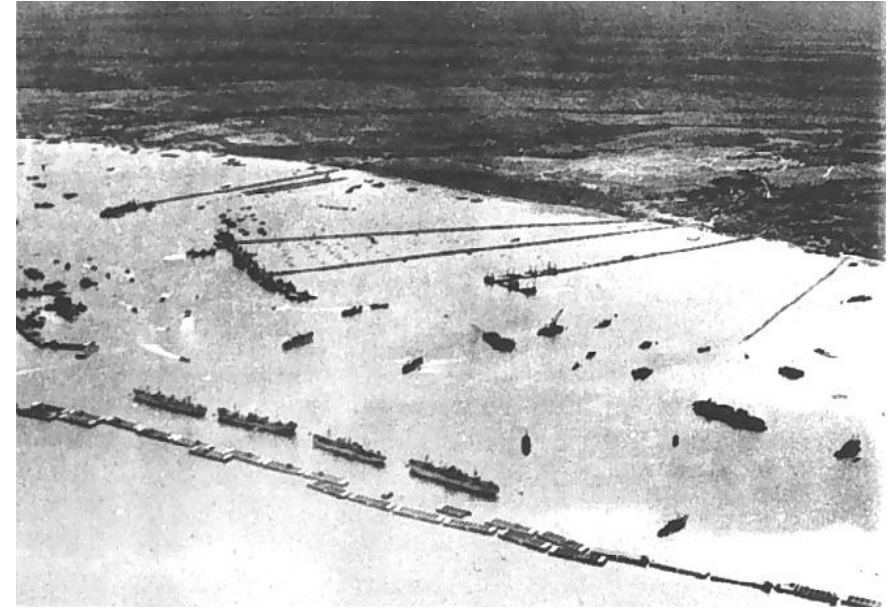
Kinds of Failure



- **Project management failure:** *Failure of the project to meet time and cost targets* (the project overran targets according to the project schedule and budget).
 - Often, however, despite the overruns, the project end item or solution is still considered a success because most stakeholders—customers, users, the market, or the community—got what they expected, or more. *The end-result performs well.*
- **Project failure:** *Failure of the project end-item to provide stakeholders with what was promise or expected.* The project did not meet performance requirements.
- For everyone, *the best scenario is* both **project management success** (time and cost) and **project success** (outcome performance)
 - *The worst scenario is project failure:* stakeholders are stuck with something that does not meet their requirements or expectations—even if it was delivered on time and on budget.

What *is* a “Project?”

- Goal-oriented
 - Aims at a specific end result or deliverables
- Somewhat unique
 - Non-routine
- Cross-functional
 - Cross-disciplinary
 - Cross-organizational
- Somewhat unfamiliar and risky
 - Involves something new or different
- Time- and resource-constrained
 - Temporary; has target completion date and target cost
- Follows logical sequence or progression of phases or stages



Mulberry harbor example

What is “Project Management?”

Cynical Definition



“Project management is the art of creating the illusion that any outcome is the result of a series of predetermined, deliberate acts when in fact it was dumb luck!”

What *is* “Project Management?”

Simple Definition



Management to accomplish project goals

What is “Project Management?”

Longer Definition



Management to

- Define and execute everything necessary to complete a **complex system** of tasks
- Achieve project end results that might be **unique and unfamiliar**
- And do it
 - by target **completion date**
 - with **constrained resources**
 - with an organization that is **cross-functional** and **newly-formed**

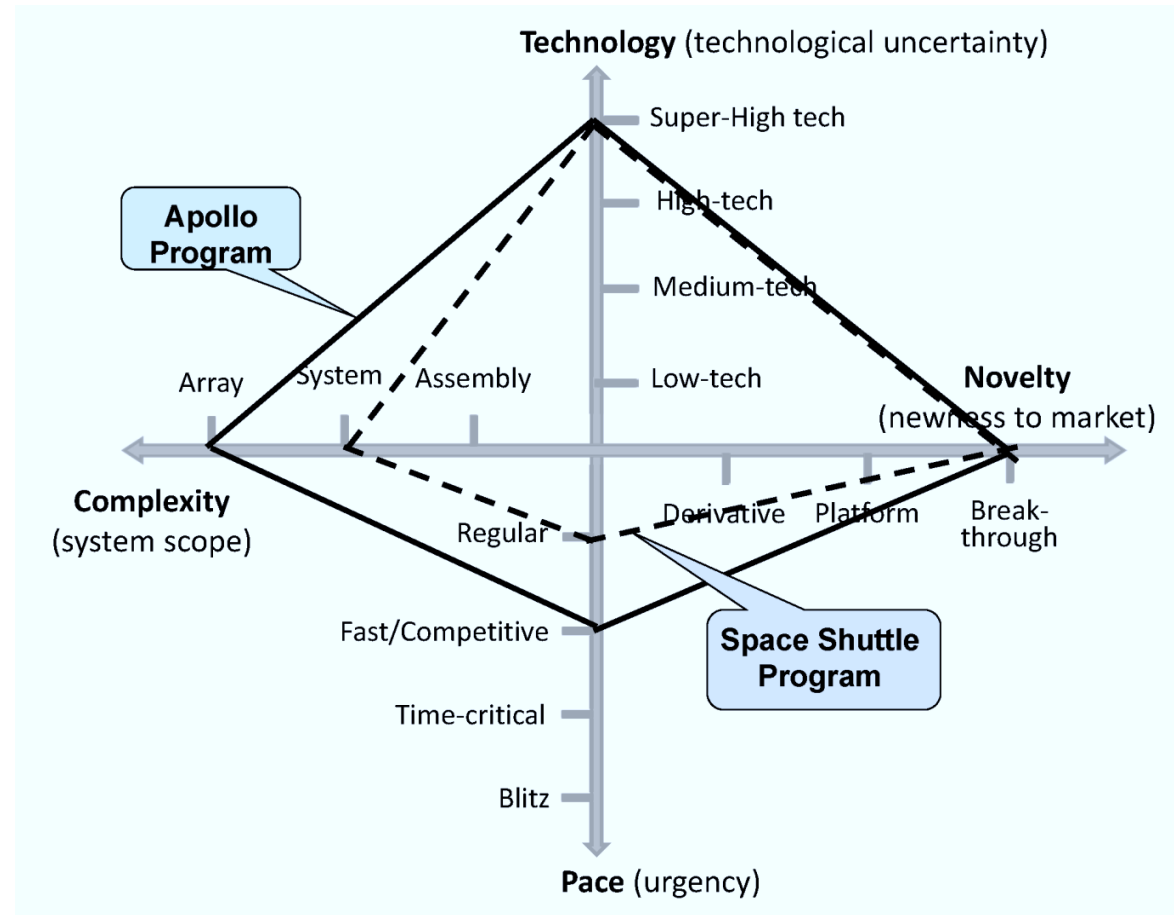
Projects are not all the same



They differ in:

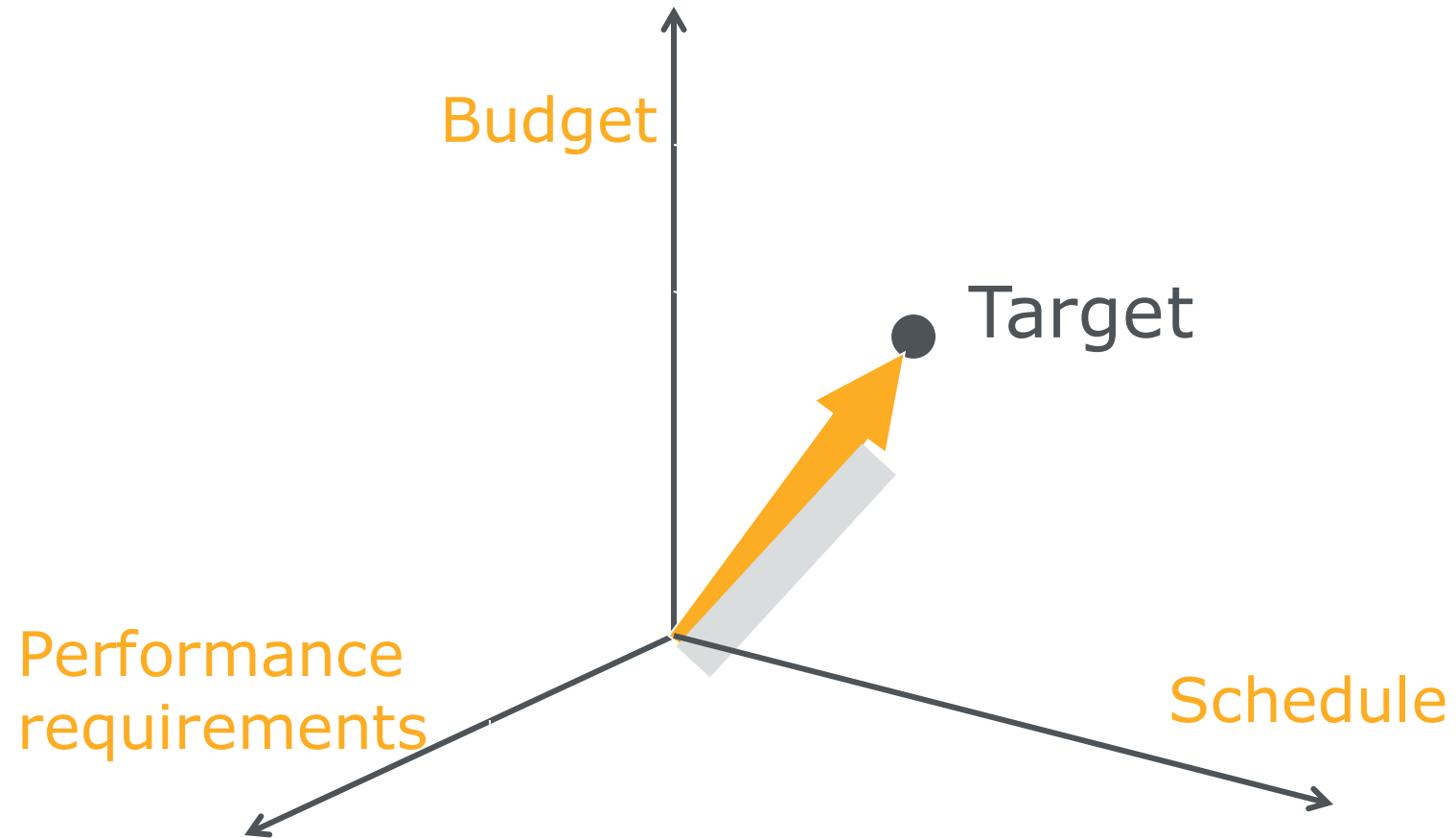
- Novelty
- Technology
- Complexity (scope)
- Pace

Projects are not all the same

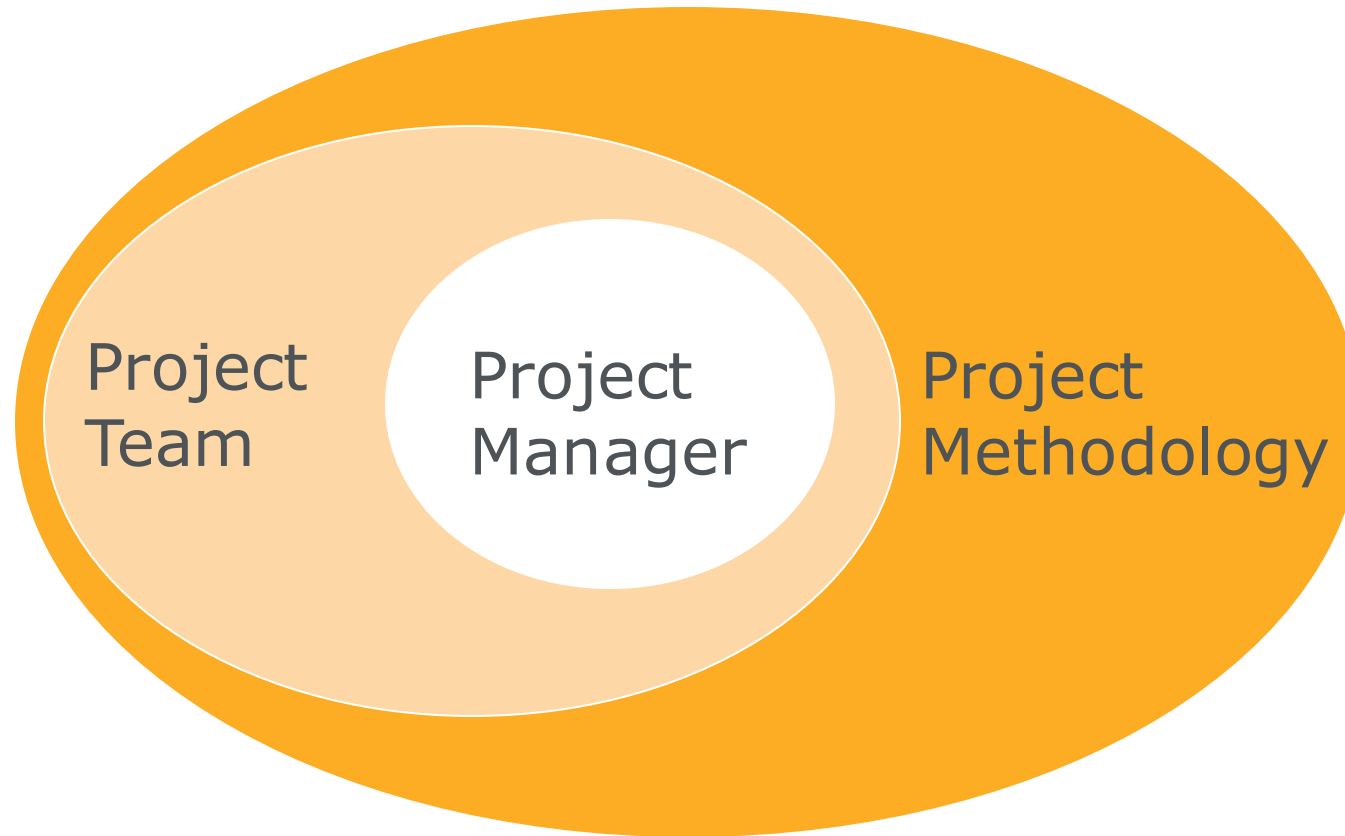


Shenhar & Dvir's NTCP Diamond model contrasting the Apollo and Space Shuttle programs.

“Three Dimensions” of Projects



Key Elements of Project Management



The Person, the Team, the Methodology



- The *project manager* — overall responsibility to plan, direct, and integrate the efforts of stakeholders to achieve the project goal.
- A project is a *team* effort, and project management means bringing individuals and groups together to form the team and directing them toward the common goal.
- Work is typically performed in phases according to a “*project management methodology*” that provides for integrative planning and control.



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QUESTIONS?